CASE STUDY | BUILT AT TATA CONSULTANCY SERVICES

ChronoSync

AI-Powered Attendance Reconciliation Engine

Turns messy learner attendance corrections into an upload-ready file, automatically.

30 min manual work per run → **30 sec** with ChronoSync

60x faster PER RUN

1. **Messy Excel form** — mixed formats, partial rows
2. **Validate + map IDs** — per-learner checks
3. **Normalize times** — MM/DD/YYYY HH:MM
4. **Upload-ready file** — plus validation report

40 learners in sample

4 program formats (1, 2, 4, 5 days)

59 missed-time instances

3 outputs: file, report, counts

TECH: Prompt Engineering · Microsoft Copilot · Claude · Python (pandas)

Sample data only: all names, emails and IDs are fictional. No client data included. Full workbook on the following pages.

ChronoSync | AI-Powered Attendance Reconciliation Engine

Built at Tata Consultancy Services for a client's L&D attendance correction process

Area	Details
The problem	Learners submit attendance corrections in an Excel form with messy entries (mixed time formats, several missed periods in one cell, partial or comment-only rows). Turning these into an upload-ready missed time file was manual and repetitive.
What the tool does	Reads the correction form, validates each entry, maps learners to User IDs, normalizes times to MM/DD/YYYY HH:MM, and fills the missed time template. It also produces a per-learner validation report and summary counts.
Scale shown here	Sample data covers 40 learners in four program formats (1, 2, 4 and 5 days), with up to three missed periods per learner per day, so it shows the tool handling many learners, many days, and multiple breaks.
Inputs	1) Attendance Corrections Form 2) Learner Email to User ID mapping file 3) Missed Time template
Outputs	1) Populated Missed Time file (drop-in ready for upload) 2) Validation report per learner 3) Summary counts, plus the analysis in this workbook
Tech	Prompt Engineering, Microsoft Copilot, Claude, Python (pandas)
Impact	Processing time cut from about 30 minutes of manual work to about 30 seconds per run (see Impact sheet).

Sample data scope (live counts)

Measure	Value

Learners enrolled	40
Form rows received	59
Missed-time instances extracted	59
Learners with missed time	23
Learners with multiple missed occurrences	14

Note: all names, emails, and IDs in this workbook are fictional sample data. No client data is included.

ChronoSync Dashboard: Attendance Correction Analytics

Sample dataset: 40 learners, 4 program formats (1, 2, 4 and 5 days), multiple missed periods per day. Fictional data.

Form rows received	Rows accepted (valid)	Missed instances found	Learners with missed time	Total missed hours	Avg minutes per instance
59	64%	59	23	38.8	39.4

Key insights (from the sample data)

1. Scale: 59 form rows from 40 enrolled learners (38 usable rows) produced 59 missed-time instances across 1, 2, 4 and 5-day programs.

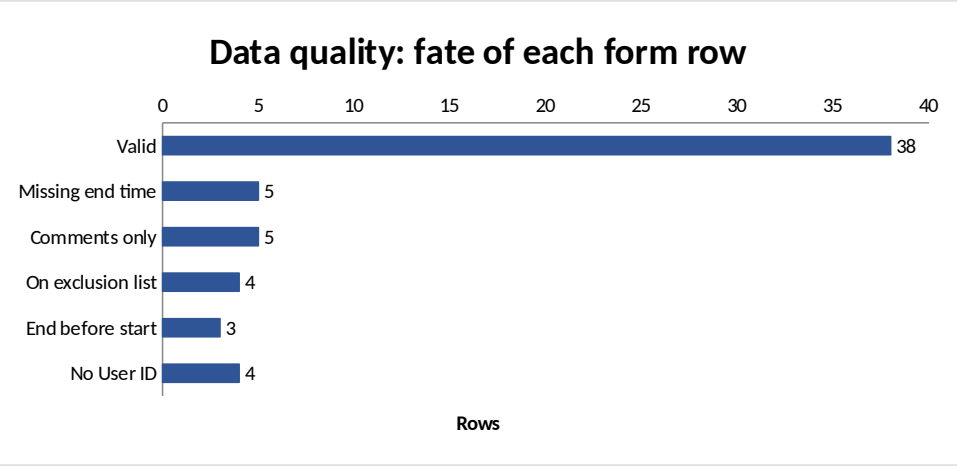
2. Multi-day effect: instances per enrolled learner are 0.42 on Day 1 and 0.80 on Day 5; the highest rate is on Day 5 (0.80) and the lowest on Day 2 (0.33).

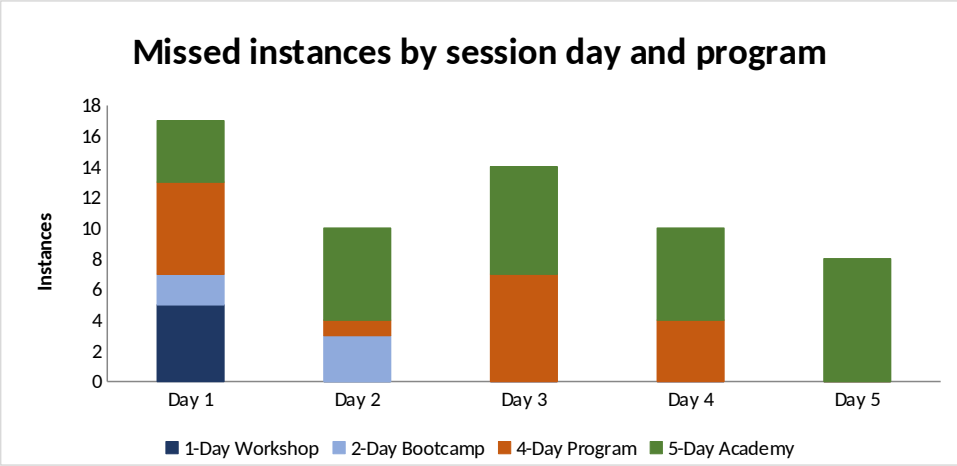
3. Timing: the most common start hour is 09:00, and 51% of missed periods start at 13:00 or later.

4. Concentration: the top 5 of 23 affected learners account for 45% of all missed minutes, so a small group drives a large share of the volume.

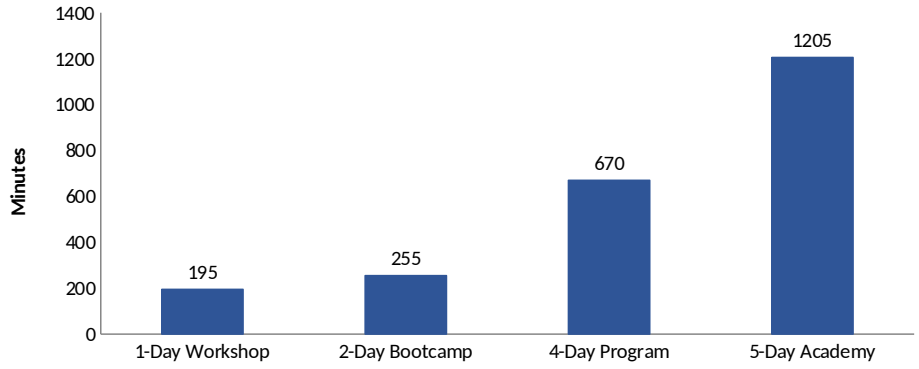
5. Data quality: 38 of 59 rows (64%) were usable; the most common rejection was 'comments only' (5 rows). 16 valid rows held more than one missed period in a single day.

Suggested actions: add a catch-up window after the peak hour, send check-ins from mid-program days, and validate the form at source (required end time, 24-hour format) to cut rejected rows.

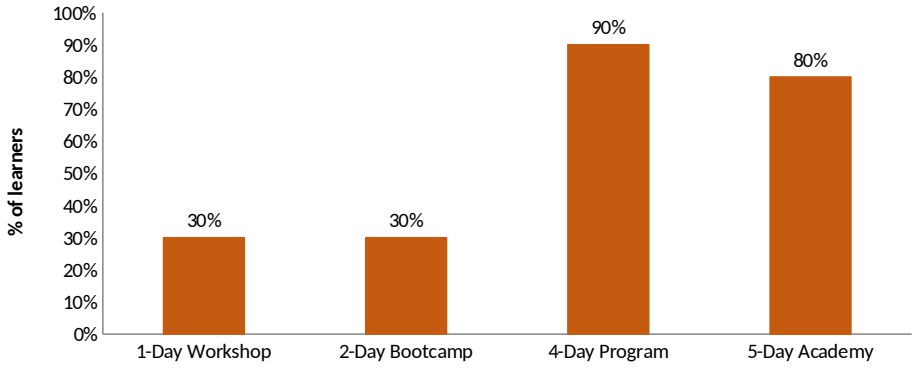




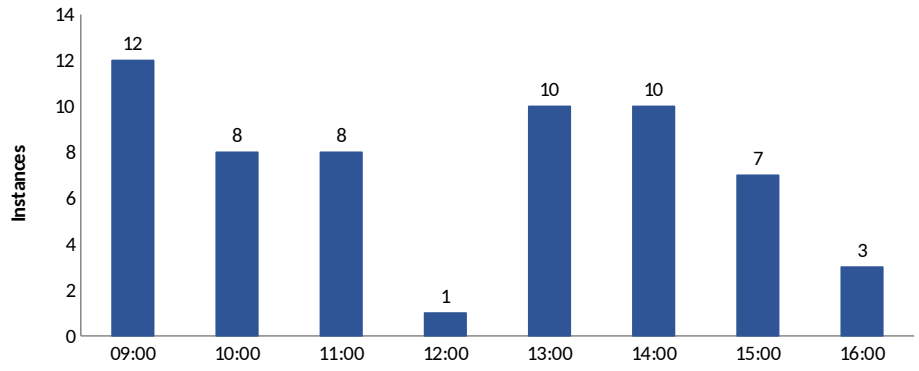
Total missed minutes by program



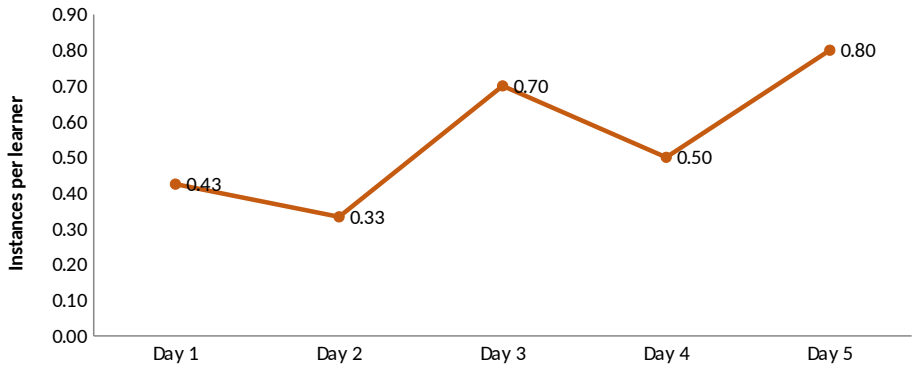
% of enrolled learners with missed time

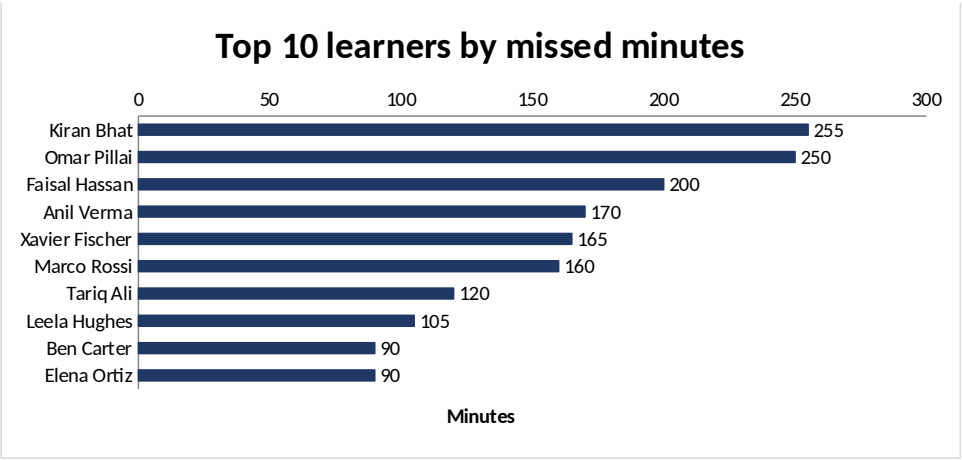


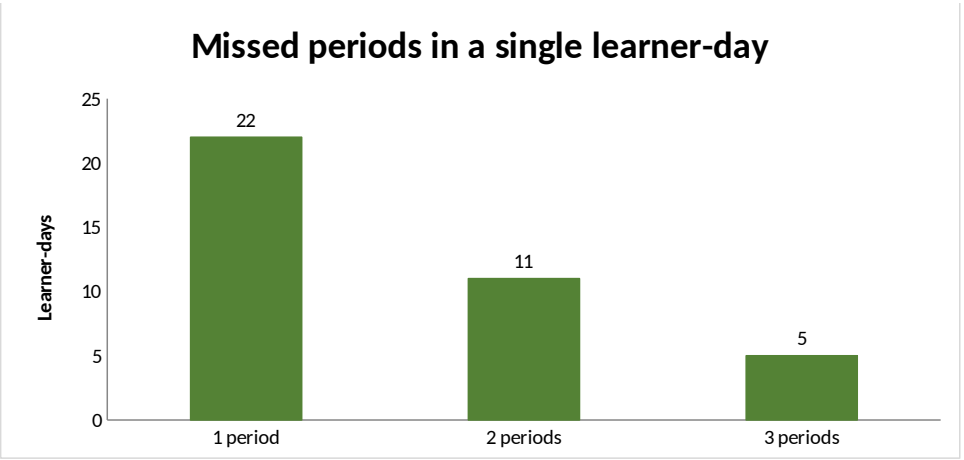
When missed time starts (hour of day)



Missed instances per enrolled learner, by day




Top 10 learners by missed minutes
0
50
100
150
200
250
300
Kiran Bhat
255
Omar Pillai
250
Faisal Hassan
200
Anil Verma
170
Xavier Fischer
165
Marco Rossi
160
Tariq Ali
120
Leela Hughes
105
Ben Carter
90
Elena Ortiz
90
Minutes


Missed periods in a single learner-day
Learner-days
25
20
15
10
5
0
22
11
5
1 period
2 periods
3 periods

Analysis tables

All figures are formulas on the sample data; charts on the Dashboard read from here.

A. Program summary

Program	Days	Enrolled learners	Learners with missed time	% learners affected	Missed instances	Missed minutes	Avg minutes per instance	Avg instances per affected learner
1-Day Workshop	1	10	3	30%	5	195	39.0	1.67
2-Day Bootcamp	2	10	3	30%	5	255	51.0	1.67
4-Day Program	4	10	9	90%	18	670	37.2	2.00
5-Day Academy	5	10	8	80%	31	1205	38.9	3.88
Total		**40**	**23**	**57%**	**59**	**2325**	**39.4**	**2.57**

B. Missed time by session day

Day	Day #	1-Day Workshop	2-Day Bootcamp	4-Day Program	5-Day Academy	Total instances	Learners in session that day	Instances per learner	Missed minutes	Avg minutes per instance
Day 1	1	5	2	6	4	17	40	0.43	685	40.3
Day 2	2	0	3	1	6	10	30	0.33	370	37.0
Day 3	3	0	0	7	7	14	20	0.70	620	44.3
Day 4	4	0	0	4	6	10	20	0.50	405	40.5
Day 5	5	0	0	0	8	8	10	0.80	245	30.6

C. Missed time by start hour

Start hour	Hour	Missed instances	Missed minutes
09:00	9	12	405
10:00	10	8	365
11:00	11	8	300
12:00	12	1	45
13:00	13	10	480
14:00	14	10	475
15:00	15	7	185
16:00	16	3	70

D. Top 10 learners by missed minutes

Rank	Learner	Program	Missed minutes	Missed instances
1	Kiran Bhat	5-Day Academy	255	8
2	Omar Pillai	5-Day Academy	250	5
3	Faisal Hassan	5-Day Academy	200	5
4	Anil Verma	4-Day Program	170	4
5	Xavier Fischer	4-Day Program	165	3
6	Marco Rossi	5-Day Academy	160	3

7	Tariq Ali	2-Day Bootcamp	120	3
8	Leela Hughes	5-Day Academy	105	5
9	Ben Carter	1-Day Workshop	90	2
10	Elena Ortiz	4-Day Program	90	3
Top 10 share of all missed minutes			**69%**	

E. Data quality: what happens to each form row

Outcome	Outcome text in input	Rows	% of rows
Valid	Valid	38	64%
Missing end time	Excluded: end time missing	5	8%
Comments only	Excluded: comments only	5	8%
On exclusion list	Excluded: on exclusion list	4	7%
End before start	Excluded: end before start	3	5%
No User ID	Excluded: no User ID found	4	7%
Total rows received		**59**	**100%**

F. Missed periods per learner-day (valid rows)

Periods in one day	Count	Learner-days	% of valid learner-days
1 period	1	22	58%
2 periods	2	11	29%
3 periods	3	5	13%

How it works: pipeline steps

Ingestion, validation, mapping, output, analysis

Step	Stage	What happens	Rule applied	Tool
1	Ingestion	Reads the corrections workbook and auto-detects the header row	Header row found by matching column names	Python (pandas)
2	Cleaning	Cleans learner emails and splits multi-entry time cells	Trim, lowercase, remove mailto/brackets; split on new line or comma	Python / Prompt
3	Filtering	Keeps only complete missed-time records	Both start and end required; partial and comment-only rows dropped; excluded emails dropped	Prompt (Copilot, Claude)
4	Time normalization	Converts 9am, 1:06 PM, 13:06 etc. into one standard format	Output MM/DD/YYYY HH:MM, 24-hour, no AM/PM	Python / Prompt
5	Time validation	Removes impossible time ranges	End time before or equal to start is invalid; no next-day inference	Python / Prompt
6	User ID mapping	Joins learner email to User ID	Unmapped learners are excluded; User IDs are never guessed	Python (pandas)
7	Output	Fills only the required template columns and builds the validation report	Template structure preserved; other columns left blank	Python / Prompt
8	Analysis	Summarizes missed time by program, session day, hour, and learner	Counts and minutes aggregated from the processed instances	Excel formulas

Sessions in the sample data

Four program formats: 1, 2, 4 and 5 days (fictional)

Session ID	Program	Days	Start date	End date	Enrolled learners
S-101	1-Day Workshop	1	10/06/2025	10/06/2025	10
S-102	2-Day Bootcamp	2	10/07/2025	10/08/2025	10
S-103	4-Day Program	4	10/13/2025	10/16/2025	10
S-104	5-Day Academy	5	10/20/2025	10/24/2025	10

Enrolled learners (blue) are sample inputs. All sessions run on consecutive weekdays.

Sample User ID mapping file

Learner Email to User ID (fictional). Some learners are intentionally missing.

Learner Email	User ID
asha.rao@example.com	U1001
ben.carter@example.com	U1002
chitra.nair@example.com	U1003
dev.patel@example.com	U1004
farah.khan@example.com	U1005
gita.menon@example.com	U1006
hiro.tanaka@example.com	U1007
isha.sharma@example.com	U1008
jon.brooks@example.com	U1009
kavya.iyer@example.com	U1010
liam.murphy@example.com	U1011
meera.das@example.com	U1012
nikhil.gupta@example.com	U1013
olivia.reed@example.com	U1014
quinn.lopez@example.com	U1015
rahul.mehta@example.com	U1016
sara.cole@example.com	U1017
tariq.ali@example.com	U1018
uma.joshi@example.com	U1019
vikram.bose@example.com	U1020
wen.chen@example.com	U1021
xavier.fischer@example.com	U1022
yara.ahmed@example.com	U1023
zane.kapoor@example.com	U1024
anil.verma@example.com	U1025
bianca.silva@example.com	U1026
carlos.nguyen@example.com	U1027
divya.reddy@example.com	U1028
elena.ortiz@example.com	U1029
faisal.hassan@example.com	U1030

grace.park@example.com	U1031
harsh.malik@example.com	U1032
ines.costa@example.com	U1033
kiran.bhat@example.com	U1034
leela.hughes@example.com	U1035
marco.rossi@example.com	U1036
nadia.ansari@example.com	U1037
omar.pillai@example.com	U1038

Learner roster with Session Learning Object IDs

All 40 enrolled learners. IDs are fictional random placeholders (24+ characters) for this sample.

Learner Name	Learner Email	User ID	Session ID	Session Learning Object ID	ID length
Asha Rao	asha.rao@example.com	U1001	S-101	SLO-C48TNNINDEHNOWTC61AI22OQ	28
Ben Carter	ben.carter@example.com	U1002	S-101	SLO-6VTHIB2TN9QJ3U3LQ2QY1UXV	28
Chitra Nair	chitra.nair@example.com	U1003	S-101	SLO-LWIITL49KCF3TU6A69TVTF73SL	30
Dev Patel	dev.patel@example.com	U1004	S-101	SLO-SUYHHFPU1JAUHFGY0J6E	24
Farah Khan	farah.khan@example.com	U1005	S-101	SLO-LNO5VQG7FYLZ5XUA106MSL89F	29
Gita Menon	gita.menon@example.com	U1006	S-101	SLO-SWWPL2886F5M0RGSCGM2C37EY	29
Hiro Tanaka	hiro.tanaka@example.com	U1007	S-101	SLO-81IJJXXUP1UXHJWWXM108VTZ	28
Isha Sharma	isha.sharma@example.com	U1008	S-101	SLO-W33OMBD4K367PT6JMN1A4TQLM	29
Jon Brooks	jon.brooks@example.com	U1009	S-101	SLO-ERZ1KJAAZKPG06TVX4F3JX	26
Kavya Iyer	kavya.iyer@example.com	U1010	S-101	SLO-QGI2D0KEJJC3TNWHB6E5KA6	27
Liam Murphy	liam.murphy@example.com	U1011	S-102	SLO-AMFK2A87YFF1W7DB19FE1	25
Meera Das	meera.das@example.com	U1012	S-102	SLO-WEUOILTWKHL2FQNBGEEG1	25
Nikhil Gupta	nikhil.gupta@example.com	U1013	S-102	SLO-M6EZHSQVE2LEEG2EXMVXJ40Y5X	30
Olivia Reed	olivia.reed@example.com	U1014	S-102	SLO-L8Q660HBXTS5W66JWE7ZVB	26
Priya Singh	priya.singh@example.com	Not mapped	S-102	SLO-L23F7L0PGRK1GGVWCBZWFD1V	28
Quinn Lopez	quinn.lopez@example.com	U1015	S-102	SLO-S2K8JYTD4IPUIFQ6YY8BJJ2	27
Rahul Mehta	rahul.mehta@example.com	U1016	S-102	SLO-K8BD342MDXRTRPEYSUEKO	25
Sara Cole	sara.cole@example.com	U1017	S-102	SLO-MNQ2YAO6VYOL74N8VP7HBJS8	28
Tariq Ali	tariq.ali@example.com	U1018	S-102	SLO-4XB54CJ6QBM2ZEIO7OT9OTG	27
Uma Joshi	uma.joshi@example.com	U1019	S-102	SLO-RJ971SNOPYTBYB1FNA3TELB6	28
Vikram Bose	vikram.bose@example.com	U1020	S-103	SLO-Z6TM2KJ9DKVWMK7GLRYMW5	26
Wen Chen	wen.chen@example.com	U1021	S-103	SLO-RM17MVCIXVUCZSBVAAGV	24
Xavier Fischer	xavier.fischer@example.com	U1022	S-103	SLO-6GPBIRMAN2HPD7E20NPVJE4	27
Yara Ahmed	yara.ahmed@example.com	U1023	S-103	SLO-46PSDTQX1EUG0O0VQTWFKK19	28
Zane Kapoor	zane.kapoor@example.com	U1024	S-103	SLO-6N58EIMC41P123ADUE34NPZ5A	29
Anil Verma	anil.verma@example.com	U1025	S-103	SLO-RA69DX9HROTFEIB2P89A	24
Bianca Silva	bianca.silva@example.com	U1026	S-103	SLO-OKUWGD9D4WDIPON5UK28	24
Carlos Nguyen	carlos.nguyen@example.com	U1027	S-103	SLO-Z1JF2RPF8W0XQ25KF659	24
Divya Reddy	divya.reddy@example.com	U1028	S-103	SLO-9ZMWZTOF9F0HRKC8FAZLG0BWZ0	30
Elena Ortiz	elena.ortiz@example.com	U1029	S-103	SLO-QIJ3GVGQ8ST8FRF136RYXL9L	28

Faisal Hassan	faisal.hassan@example.com	U1030	S-104	SLO-OB918VMHNUEFD8QLKA4GDOJD	28
Grace Park	grace.park@example.com	U1031	S-104	SLO-0OWZ6RUEELX40CXY5HKN6BBEL	29
Harsh Malik	harsh.malik@example.com	U1032	S-104	SLO-BWVK621ZK3QLU54J9XY4VAZ3N2	30
Ines Costa	ines.costa@example.com	U1033	S-104	SLO-CGRJR1VP7UVUE3FKSGSW6	25
Jamal Khanna	jamal.khanna@example.com	Not mapped	S-104	SLO-CCSKA21VY21JWRYNBQGO	24
Kiran Bhat	kiran.bhat@example.com	U1034	S-104	SLO-8H70VTYKD3MTTMZCW8W5YF	26
Leela Hughes	leela.hughes@example.com	U1035	S-104	SLO-M61KA3YETXSWWO2QM8IW1WUR	28
Marco Rossi	marco.rossi@example.com	U1036	S-104	SLO-1NTR2AJOJDILO4L4YLJPZJ7VB	29
Nadia Ansari	nadia.ansari@example.com	U1037	S-104	SLO-ZB1J585Z0MRVVA3WWWSMN	25
Omar Pillai	omar.pillai@example.com	U1038	S-104	SLO-VSIXVAUKOVQNW8DCCEIGKHY	27

Minimum ID length in this roster: **24**

In the real tool, Session Learning Object IDs come from the template or source system and are never generated or guessed.

Sample input: Attendance Corrections Form (messy, as received)

Fictional data. Mixed time formats, several breaks in one cell, and invalid rows. Outcome shows what the tool does with each row.

Session ID	Learner Name	Learner Email Address	Date (MM/DD/YYYY)	Missed Time Start	Missed Time End	Comments	Mapped User ID?	Outcome	Missed periods in cell
S-101	Asha Rao	asha.rao@example.com	10/06/2025			Joined late, will update	Yes	Excluded: comments only	0
S-101	Ben Carter	BEN.CARTER@EXAMPLE.COM	10/06/2025	9:40 11:00	9:55 12:15	Platform glitch	Yes	Valid	2
S-101	Chitra Nair	chitra.nair@example.com	10/06/2025	10:00		End time not noted	Yes	Excluded: end time missing	1
S-101	Dev Patel	dev.patel@example.com	10/06/2025	9:45	10:00		Yes	Valid	1
S-101	Farah Khan	farah.khan@example.com	10/06/2025			Left early for another meeting	Yes	Excluded: comments only	0
S-101	Jon Brooks	jon.brooks@example.com	10/06/2025	10:10 AM 3:10 PM	11:25 AM 3:25 PM	Audio problem	Yes	Valid	2
S-102	Priya Singh	PRIYA.SINGH@EXAMPLE.COM	10/07/2025	2:10pm	3:10pm		No	Excluded: no User ID found	1
S-102	Quinn Lopez	quinn.lopez@example.com	10/07/2025	2pm	1pm	Typed times wrongly	Yes	Excluded: end before start	1
S-102	Tariq Ali	tariq.ali@example.com	10/07/2025	10:30 14:50	11:10 15:50	Rejoined after break	Yes	Valid	2
S-102	Liam Murphy	liam.murphy@example.com	10/08/2025	10:50am	12:05pm		Yes	Valid	1
S-102	Olivia Reed	olivia.reed@example.com	10/08/2025	1:25pm	2:25pm	Client call	Yes	Valid	1
S-102	Priya Singh	priya.singh@example.com	10/08/2025	2:25 PM	3:05 PM		No	Excluded: no User ID found	1
S-102	Tariq Ali	tariq.ali@example.com	10/08/2025	9:20 AM	9:40 AM	Platform glitch	Yes	Valid	1
S-103	Bianca Silva	bianca.silva@example.com	10/13/2025	2pm	2:30pm		Yes	Valid	1
S-103	Elena Ortiz	elena.ortiz@example.com	10/13/2025	1:10 PM 3:55 PM	1:55 PM 4:20 PM	Platform glitch	Yes	Valid	2
S-103	Excluded User 1	excluded.user1@example.com	10/13/2025	10:00	10:30		No	Excluded: on exclusion list	1
S-103	Vikram Bose	vikram.bose@example.com	10/13/2025	10:10 AM	10:55 AM	Platform glitch	Yes	Valid	1
S-103	Yara Ahmed	yara.ahmed@example.com	10/13/2025	9:45 13:25	10:05 13:35		Yes	Valid	2
S-103	Excluded User 1	excluded.user1@example.com	10/14/2025	10:00	10:30		No	Excluded: on exclusion list	1
S-103	Wen Chen	wen.chen@example.com	10/14/2025	9:40 AM	10:20 AM	Laptop issue	Yes	Valid	1
S-103	Anil Verma	anil.verma@example.com	10/15/2025	11:55 13:10 15:40	13:10 13:50 16:10		Yes	Valid	3
S-103	Wen Chen	wen.chen@example.com	10/15/2025	2pm	1pm	Typed times wrongly	Yes	Excluded: end before start	1
S-103	Xavier Fischer	Xavier.fischer@example.com	10/15/2025	9:50 11:30 14:50	10:35 12:15 16:05	Audio problem	Yes	Valid	3
S-103	Zane Kapoor	zane.kapoor@example.com	10/15/2025	11:10 AM	11:25 AM	Connection lost	Yes	Valid	1
S-103	Anil Verma	anil.verma@example.com	10/16/2025	3:15pm	3:40pm	Audio problem	Yes	Valid	1
S-103	Carlos Nguyen	carlos.nguyen@example.com	10/16/2025	10:15 AM	11:15 AM		Yes	Valid	1
S-103	Elena Ortiz	elena.ortiz@example.com	10/16/2025	4 PM	4:20 PM		Yes	Valid	1

S-103	Vikram Bose	vikram.bose@example.com	10/16/2025	9:10 AM	9:35 AM		Yes	Valid	1
S-103	Zane Kapoor	zane.kapoor@example.com	10/16/2025			Joined late, will update	Yes	Excluded: comments only	0
S-104	Excluded User 2	excluded.user2@example.com	10/20/2025	10:00	10:30		No	Excluded: on exclusion list	1
S-104	Faisal Hassan	faisal.hassan@example.com	10/20/2025	9:40 13:10	10:00 13:55	Audio problem	Yes	Valid	2
S-104	Grace Park	grace.park@example.com	10/20/2025	13:05	14:20		Yes	Valid	1
S-104	Harsh Malik	harsh.malik@example.com	10/20/2025	9:05am	10:20am	Client call	Yes	Valid	1
S-104	Kiran Bhat	kiran.bhat@example.com	10/20/2025			Will share exact times later	Yes	Excluded: comments only	0
S-104	Leela Hughes	leela.hughes@example.com	10/20/2025	13:00		End time not noted	Yes	Excluded: end time missing	1
S-104	Excluded User 2	excluded.user2@example.com	10/21/2025	10:00	10:30		No	Excluded: on exclusion list	1
S-104	Faisal Hassan	faisal.hassan@example.com	10/21/2025	1:45pm	2:25pm	Laptop issue	Yes	Valid	1
S-104	Grace Park	grace.park@example.com	10/21/2025	15:30		End time not noted	Yes	Excluded: end time missing	1
S-104	Harsh Malik	harsh.malik@example.com	10/21/2025	9:00		End time not noted	Yes	Excluded: end time missing	1
S-104	Kiran Bhat	kiran.bhat@example.com	10/21/2025	11:35am 2:40pm 4:15pm	11:55am 3:25pm 4:35pm	Rejoined after break	Yes	Valid	3
S-104	Nadia Ansari	nadia.ansari@example.com	10/21/2025	10:55am	11:15am	Network drop	Yes	Valid	1
S-104	Omar Pillai	omar.pillai@example.com	10/21/2025	10am	10:30am		Yes	Valid	1
S-104	Faisal Hassan	faisal.hassan@example.com	10/22/2025	11:40 13:40	12:00 14:55	Laptop issue	Yes	Valid	2
S-104	Grace Park	grace.park@example.com	10/22/2025			Left early for another meeting	Yes	Excluded: comments only	0
S-104	Kiran Bhat	kiran.bhat@example.com	10/22/2025	9:40am 12pm	10:40am 12:45pm	Connection lost	Yes	Valid	2
S-104	Leela Hughes	leela.hughes@example.com	10/22/2025	2:30pm	2:45pm		Yes	Valid	1
S-104	Marco Rossi	marco.rossi@example.com	10/22/2025	14:20	15:00	Network drop	Yes	Valid	1
S-104	Omar Pillai	OMAR.PILLAI@EXAMPLE.COM	10/22/2025	15:15	15:55	Platform glitch	Yes	Valid	1
S-104	Grace Park	grace.park@example.com	10/23/2025	2pm	1pm	Typed times wrongly	Yes	Excluded: end before start	1
S-104	Jamal Khanna	jamal.khanna@example.com	10/23/2025	4:35pm	4:55pm		No	Excluded: no User ID found	1
S-104	Leela Hughes	leela.hughes@example.com	10/23/2025	1:15pm	1:30pm		Yes	Valid	1
S-104	Marco Rossi	marco.rossi@example.com	10/23/2025	1:15 PM 2:50 PM	2:30 PM 3:35 PM	Platform glitch	Yes	Valid	2
S-104	Nadia Ansari	nadia.ansari@example.com	10/23/2025	9:50 AM 3:30 PM	10:35 AM 3:50 PM	Connection lost	Yes	Valid	2
S-104	Omar Pillai	omar.pillai@example.com	10/23/2025	14:15	15:30	Client call	Yes	Valid	1
S-104	Ines Costa	ines.costa@example.com	10/24/2025	10:30		End time not noted	Yes	Excluded: end time missing	1
S-104	Jamal Khanna	jamal.khanna@example.com	10/24/2025	12:15 14:10	12:45 14:50	Laptop issue	No	Excluded: no User ID found	2
S-104	Kiran Bhat	kiran.bhat@example.com	10/24/2025	10am 2:05pm 3:50pm	10:20am 2:20pm 4:20pm		Yes	Valid	3

S-104	Leela Hughes	Leela.hughes@example.com	10/24/2025	9:55 11:20 16:10	10:20 11:40 16:40	Connection lost	Yes	Valid	3
S-104	Omar Pillai	omar.pillai@example.com	10/24/2025	11:20 14:05	11:50 15:20		Yes	Valid	2

Processed missed-time instances (one row per missed period)

Output of validation, mapping and normalization; the analysis sheets read from this table.

Session ID	Learner Name	Learner Email	User ID	Date	Start Time	End Time	Program	Session Day #	Duration (min)	Start Hour
S-101	Ben Carter	ben.carter@example.com	U1002	10/06/2025	09:40	09:55	1-Day Workshop	1	15	9
S-101	Ben Carter	ben.carter@example.com	U1002	10/06/2025	11:00	12:15	1-Day Workshop	1	75	11
S-101	Dev Patel	dev.patel@example.com	U1004	10/06/2025	09:45	10:00	1-Day Workshop	1	15	9
S-101	Jon Brooks	jon.brooks@example.com	U1009	10/06/2025	10:10	11:25	1-Day Workshop	1	75	10
S-101	Jon Brooks	jon.brooks@example.com	U1009	10/06/2025	15:10	15:25	1-Day Workshop	1	15	15
S-102	Tariq Ali	tariq.ali@example.com	U1018	10/07/2025	10:30	11:10	2-Day Bootcamp	1	40	10
S-102	Tariq Ali	tariq.ali@example.com	U1018	10/07/2025	14:50	15:50	2-Day Bootcamp	1	60	14
S-102	Liam Murphy	liam.murphy@example.com	U1011	10/08/2025	10:50	12:05	2-Day Bootcamp	2	75	10
S-102	Olivia Reed	olivia.reed@example.com	U1014	10/08/2025	13:25	14:25	2-Day Bootcamp	2	60	13
S-102	Tariq Ali	tariq.ali@example.com	U1018	10/08/2025	09:20	09:40	2-Day Bootcamp	2	20	9
S-103	Bianca Silva	bianca.silva@example.com	U1026	10/13/2025	14:00	14:30	4-Day Program	1	30	14
S-103	Elena Ortiz	elena.ortiz@example.com	U1029	10/13/2025	13:10	13:55	4-Day Program	1	45	13
S-103	Elena Ortiz	elena.ortiz@example.com	U1029	10/13/2025	15:55	16:20	4-Day Program	1	25	15
S-103	Vikram Bose	vikram.bose@example.com	U1020	10/13/2025	10:10	10:55	4-Day Program	1	45	10
S-103	Yara Ahmed	yara.ahmed@example.com	U1023	10/13/2025	09:45	10:05	4-Day Program	1	20	9
S-103	Yara Ahmed	yara.ahmed@example.com	U1023	10/13/2025	13:25	13:35	4-Day Program	1	10	13
S-103	Wen Chen	wen.chen@example.com	U1021	10/14/2025	09:40	10:20	4-Day Program	2	40	9
S-103	Anil Verma	anil.verma@example.com	U1025	10/15/2025	11:55	13:10	4-Day Program	3	75	11
S-103	Anil Verma	anil.verma@example.com	U1025	10/15/2025	13:10	13:50	4-Day Program	3	40	13
S-103	Anil Verma	anil.verma@example.com	U1025	10/15/2025	15:40	16:10	4-Day Program	3	30	15
S-103	Xavier Fischer	xavier.fischer@example.com	U1022	10/15/2025	09:50	10:35	4-Day Program	3	45	9
S-103	Xavier Fischer	xavier.fischer@example.com	U1022	10/15/2025	11:30	12:15	4-Day Program	3	45	11
S-103	Xavier Fischer	xavier.fischer@example.com	U1022	10/15/2025	14:50	16:05	4-Day Program	3	75	14
S-103	Zane Kapoor	zane.kapoor@example.com	U1024	10/15/2025	11:10	11:25	4-Day Program	3	15	11
S-103	Anil Verma	anil.verma@example.com	U1025	10/16/2025	15:15	15:40	4-Day Program	4	25	15
S-103	Carlos Nguyen	carlos.nguyen@example.com	U1027	10/16/2025	10:15	11:15	4-Day Program	4	60	10
S-103	Elena Ortiz	elena.ortiz@example.com	U1029	10/16/2025	16:00	16:20	4-Day Program	4	20	16
S-103	Vikram Bose	vikram.bose@example.com	U1020	10/16/2025	09:10	09:35	4-Day Program	4	25	9
S-104	Faisal Hassan	faisal.hassan@example.com	U1030	10/20/2025	09:40	10:00	5-Day Academy	1	20	9
S-104	Faisal Hassan	faisal.hassan@example.com	U1030	10/20/2025	13:10	13:55	5-Day Academy	1	45	13
S-104	Grace Park	grace.park@example.com	U1031	10/20/2025	13:05	14:20	5-Day Academy	1	75	13
S-104	Harsh Malik	harsh.malik@example.com	U1032	10/20/2025	09:05	10:20	5-Day Academy	1	75	9
S-104	Faisal Hassan	faisal.hassan@example.com	U1030	10/21/2025	13:45	14:25	5-Day Academy	2	40	13
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/21/2025	11:35	11:55	5-Day Academy	2	20	11
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/21/2025	14:40	15:25	5-Day Academy	2	45	14

S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/21/2025	16:15	16:35	5-Day Academy	2	20	16
S-104	Nadia Ansari	nadia.ansari@example.com	U1037	10/21/2025	10:55	11:15	5-Day Academy	2	20	10
S-104	Omar Pillai	omar.pillai@example.com	U1038	10/21/2025	10:00	10:30	5-Day Academy	2	30	10
S-104	Faisal Hassan	faisal.hassan@example.com	U1030	10/22/2025	11:40	12:00	5-Day Academy	3	20	11
S-104	Faisal Hassan	faisal.hassan@example.com	U1030	10/22/2025	13:40	14:55	5-Day Academy	3	75	13
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/22/2025	09:40	10:40	5-Day Academy	3	60	9
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/22/2025	12:00	12:45	5-Day Academy	3	45	12
S-104	Leela Hughes	leela.hughes@example.com	U1035	10/22/2025	14:30	14:45	5-Day Academy	3	15	14
S-104	Marco Rossi	marco.rossi@example.com	U1036	10/22/2025	14:20	15:00	5-Day Academy	3	40	14
S-104	Omar Pillai	omar.pillai@example.com	U1038	10/22/2025	15:15	15:55	5-Day Academy	3	40	15
S-104	Leela Hughes	leela.hughes@example.com	U1035	10/23/2025	13:15	13:30	5-Day Academy	4	15	13
S-104	Marco Rossi	marco.rossi@example.com	U1036	10/23/2025	13:15	14:30	5-Day Academy	4	75	13
S-104	Marco Rossi	marco.rossi@example.com	U1036	10/23/2025	14:50	15:35	5-Day Academy	4	45	14
S-104	Nadia Ansari	nadia.ansari@example.com	U1037	10/23/2025	09:50	10:35	5-Day Academy	4	45	9
S-104	Nadia Ansari	nadia.ansari@example.com	U1037	10/23/2025	15:30	15:50	5-Day Academy	4	20	15
S-104	Omar Pillai	omar.pillai@example.com	U1038	10/23/2025	14:15	15:30	5-Day Academy	4	75	14
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/24/2025	10:00	10:20	5-Day Academy	5	20	10
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/24/2025	14:05	14:20	5-Day Academy	5	15	14
S-104	Kiran Bhat	kiran.bhat@example.com	U1034	10/24/2025	15:50	16:20	5-Day Academy	5	30	15
S-104	Leela Hughes	leela.hughes@example.com	U1035	10/24/2025	09:55	10:20	5-Day Academy	5	25	9
S-104	Leela Hughes	leela.hughes@example.com	U1035	10/24/2025	11:20	11:40	5-Day Academy	5	20	11
S-104	Leela Hughes	leela.hughes@example.com	U1035	10/24/2025	16:10	16:40	5-Day Academy	5	30	16
S-104	Omar Pillai	omar.pillai@example.com	U1038	10/24/2025	11:20	11:50	5-Day Academy	5	30	11
S-104	Omar Pillai	omar.pillai@example.com	U1038	10/24/2025	14:05	15:20	5-Day Academy	5	75	14

Sample output: populated Missed Time template (upload-ready)

User ID, Session Learning Object ID (sample placeholder, looked up from the Learner Roster) and the two missed time columns.

User ID	Session Learning Object ID	Missed Start Time	Missed End Time
U1002	SLO-6VTHIB2TN9QJ3U3LQ2QY1	10/06/2025 09:40	10/06/2025 09:55
U1002	SLO-6VTHIB2TN9QJ3U3LQ2QY1	10/06/2025 11:00	10/06/2025 12:15
U1004	SLO-SUYHHFPU1JAUHFGY0J6E	10/06/2025 09:45	10/06/2025 10:00
U1009	SLO-ERZ1KJAAZKPG06TVX4F3J	10/06/2025 10:10	10/06/2025 11:25
U1009	SLO-ERZ1KJAAZKPG06TVX4F3J	10/06/2025 15:10	10/06/2025 15:25
U1018	SLO-4XB54CJ6QBM2ZEIO7OT9O	10/07/2025 10:30	10/07/2025 11:10
U1018	SLO-4XB54CJ6QBM2ZEIO7OT9O	10/07/2025 14:50	10/07/2025 15:50
U1011	SLO-AMFK2A87YFF1W7DB19FE	10/08/2025 10:50	10/08/2025 12:05
U1014	SLO-L8Q660HBXTS5W66JWE7Z	10/08/2025 13:25	10/08/2025 14:25
U1018	SLO-4XB54CJ6QBM2ZEIO7OT9O	10/08/2025 09:20	10/08/2025 09:40
U1026	SLO-OKUWGD9D4WDIPON5UK2	10/13/2025 14:00	10/13/2025 14:30
U1029	SLO-QIJ3GVGQ8ST8FRF136RYX	10/13/2025 13:10	10/13/2025 13:55
U1029	SLO-QIJ3GVGQ8ST8FRF136RYX	10/13/2025 15:55	10/13/2025 16:20
U1020	SLO-Z6TM2KJ9DKVWMK7GLRY	10/13/2025 10:10	10/13/2025 10:55
U1023	SLO-46PSDTQX1EUG0O0VQTWF	10/13/2025 09:45	10/13/2025 10:05
U1023	SLO-46PSDTQX1EUG0O0VQTWF	10/13/2025 13:25	10/13/2025 13:35
U1021	SLO-RM17MVCIXVUCZSBVAAGV	10/14/2025 09:40	10/14/2025 10:20
U1025	SLO-RA69DX9HROTFEIB2P89A	10/15/2025 11:55	10/15/2025 13:10
U1025	SLO-RA69DX9HROTFEIB2P89A	10/15/2025 13:10	10/15/2025 13:50
U1025	SLO-RA69DX9HROTFEIB2P89A	10/15/2025 15:40	10/15/2025 16:10
U1022	SLO-6GPBIRMAN2HPD7E20NPV	10/15/2025 09:50	10/15/2025 10:35
U1022	SLO-6GPBIRMAN2HPD7E20NPV	10/15/2025 11:30	10/15/2025 12:15
U1022	SLO-6GPBIRMAN2HPD7E20NPV	10/15/2025 14:50	10/15/2025 16:05
U1024	SLO-6N58EIMC41P123ADUE34NI	10/15/2025 11:10	10/15/2025 11:25
U1025	SLO-RA69DX9HROTFEIB2P89A	10/16/2025 15:15	10/16/2025 15:40
U1027	SLO-Z1JF2RPF8W0XQ25KF659	10/16/2025 10:15	10/16/2025 11:15
U1029	SLO-QIJ3GVGQ8ST8FRF136RYX	10/16/2025 16:00	10/16/2025 16:20
U1020	SLO-Z6TM2KJ9DKVWMK7GLRY	10/16/2025 09:10	10/16/2025 09:35
U1030	SLO-OB918VMHNUEFD8QLKA4G	10/20/2025 09:40	10/20/2025 10:00
U1030	SLO-OB918VMHNUEFD8QLKA4G	10/20/2025 13:10	10/20/2025 13:55

U1031	SLO-0OWZ6RUEELX40CXY5HKN	10/20/2025 13:05	10/20/2025 14:20
U1032	SLO-BWVK621ZK3QLU54J9XY4V	10/20/2025 09:05	10/20/2025 10:20
U1030	SLO-OB918VMHNUEFD8QLKA4G	10/21/2025 13:45	10/21/2025 14:25
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/21/2025 11:35	10/21/2025 11:55
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/21/2025 14:40	10/21/2025 15:25
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/21/2025 16:15	10/21/2025 16:35
U1037	SLO-ZB1J585Z0MRVVA3WWWSM	10/21/2025 10:55	10/21/2025 11:15
U1038	SLO-VSIXVAUKOVQNW8DCCEIG	10/21/2025 10:00	10/21/2025 10:30
U1030	SLO-OB918VMHNUEFD8QLKA4G	10/22/2025 11:40	10/22/2025 12:00
U1030	SLO-OB918VMHNUEFD8QLKA4G	10/22/2025 13:40	10/22/2025 14:55
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/22/2025 09:40	10/22/2025 10:40
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/22/2025 12:00	10/22/2025 12:45
U1035	SLO-M61KA3YETXSWWO2QM8I	10/22/2025 14:30	10/22/2025 14:45
U1036	SLO-1NTR2AJOJDILO4L4YLJPZJ	10/22/2025 14:20	10/22/2025 15:00
U1038	SLO-VSIXVAUKOVQNW8DCCEIG	10/22/2025 15:15	10/22/2025 15:55
U1035	SLO-M61KA3YETXSWWO2QM8I	10/23/2025 13:15	10/23/2025 13:30
U1036	SLO-1NTR2AJOJDILO4L4YLJPZJ	10/23/2025 13:15	10/23/2025 14:30
U1036	SLO-1NTR2AJOJDILO4L4YLJPZJ	10/23/2025 14:50	10/23/2025 15:35
U1037	SLO-ZB1J585Z0MRVVA3WWWSM	10/23/2025 09:50	10/23/2025 10:35
U1037	SLO-ZB1J585Z0MRVVA3WWWSM	10/23/2025 15:30	10/23/2025 15:50
U1038	SLO-VSIXVAUKOVQNW8DCCEIG	10/23/2025 14:15	10/23/2025 15:30
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/24/2025 10:00	10/24/2025 10:20
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/24/2025 14:05	10/24/2025 14:20
U1034	SLO-8H70VTYKD3MTTMZCW8W9	10/24/2025 15:50	10/24/2025 16:20
U1035	SLO-M61KA3YETXSWWO2QM8I	10/24/2025 09:55	10/24/2025 10:20
U1035	SLO-M61KA3YETXSWWO2QM8I	10/24/2025 11:20	10/24/2025 11:40
U1035	SLO-M61KA3YETXSWWO2QM8I	10/24/2025 16:10	10/24/2025 16:40
U1038	SLO-VSIXVAUKOVQNW8DCCEIG	10/24/2025 11:20	10/24/2025 11:50
U1038	SLO-VSIXVAUKOVQNW8DCCEIG	10/24/2025 14:05	10/24/2025 15:20

Validation report (grouped by learner)

Missed times merged chronologically as comma-separated values

Measure	Value
Total learners processed	23
Total missed instances identified	59
Learners with multiple missed occurrences	14
Max instances for one learner	8

Learner Name	Learner Email	User ID	Session ID	Program	Missed Start Times	Missed End Times	Missed Instance Count	Total Missed Minutes
Ben Carter	ben.carter@example.com	U1002	S-101	1-Day Workshop	10/06/2025 09:40, 10/06/2025 11:00	10/06/2025 09:55, 10/06/2025 12:15	2	90
Dev Patel	dev.patel@example.com	U1004	S-101	1-Day Workshop	10/06/2025 09:45	10/06/2025 10:00	1	15
Jon Brooks	jon.brooks@example.com	U1009	S-101	1-Day Workshop	10/06/2025 10:10, 10/06/2025 15:10	10/06/2025 11:25, 10/06/2025 15:25	2	90
Liam Murphy	liam.murphy@example.com	U1011	S-102	2-Day Bootcamp	10/08/2025 10:50	10/08/2025 12:05	1	75
Olivia Reed	olivia.reed@example.com	U1014	S-102	2-Day Bootcamp	10/08/2025 13:25	10/08/2025 14:25	1	60
Tariq Ali	tariq.ali@example.com	U1018	S-102	2-Day Bootcamp	10/07/2025 10:30, 10/07/2025 14:50, 10/08/2025 09:20	10/07/2025 11:10, 10/07/2025 15:50, 10/08/2025 09:40	3	120
Anil Verma	anil.verma@example.com	U1025	S-103	4-Day Program	10/15/2025 11:55, 10/15/2025 13:10, 10/15/2025 15:40, 10/16/2025 15:15	10/15/2025 13:10, 10/15/2025 13:50, 10/15/2025 16:10, 10/16/2025 15:40	4	170
Bianca Silva	bianca.silva@example.com	U1026	S-103	4-Day Program	10/13/2025 14:00	10/13/2025 14:30	1	30
Carlos Nguyen	carlos.nguyen@example.com	U1027	S-103	4-Day Program	10/16/2025 10:15	10/16/2025 11:15	1	60
Elena Ortiz	elena.ortiz@example.com	U1029	S-103	4-Day Program	10/13/2025 13:10, 10/13/2025 15:55, 10/16/2025 16:00	10/13/2025 13:55, 10/13/2025 16:20, 10/16/2025 16:20	3	90
Vikram Bose	vikram.bose@example.com	U1020	S-103	4-Day Program	10/13/2025 10:10, 10/16/2025 09:10	10/13/2025 10:55, 10/16/2025 09:35	2	70
Wen Chen	wen.chen@example.com	U1021	S-103	4-Day Program	10/14/2025 09:40	10/14/2025 10:20	1	40
Xavier Fischer	xavier.fischer@example.com	U1022	S-103	4-Day Program	10/15/2025 09:50, 10/15/2025 11:30, 10/15/2025 14:50	10/15/2025 10:35, 10/15/2025 12:15, 10/15/2025 16:05	3	165
Yara Ahmed	yara.ahmed@example.com	U1023	S-103	4-Day Program	10/13/2025 09:45, 10/13/2025 13:25	10/13/2025 10:05, 10/13/2025 13:35	2	30
Zane Kapoor	zane.kapoor@example.com	U1024	S-103	4-Day Program	10/15/2025 11:10	10/15/2025 11:25	1	15
Faisal Hassan	faisal.hassan@example.com	U1030	S-104	5-Day Academy	10/20/2025 09:40, 10/20/2025 13:10, 10/21/2025 13:45, 10/22/2025 11:40, 10/22/2025 13:40	10/20/2025 10:00, 10/20/2025 13:55, 10/21/2025 14:25, 10/22/2025 12:00, 10/22/2025 14:55	5	200
Grace Park	grace.park@example.com	U1031	S-104	5-Day Academy	10/20/2025 13:05	10/20/2025 14:20	1	75
Harsh Malik	harsh.malik@example.com	U1032	S-104	5-Day Academy	10/20/2025 09:05	10/20/2025 10:20	1	75
Kiran Bhat	kiran.bhat@example.com	U1034	S-104	5-Day Academy	10/21/2025 11:35, 10/21/2025 14:40, 10/21/2025 16:15, 10/22/2025 09:40, 10/22/2025 12:00, 10/24/2025 10:00, 10/24/2025 14:05, 10/24/2025 15:50	10/21/2025 11:55, 10/21/2025 15:25, 10/21/2025 16:35, 10/22/2025 10:40, 10/22/2025 12:45, 10/24/2025 10:20, 10/24/2025 14:20, 10/24/2025 16:20	8	255
Leela Hughes	leela.hughes@example.com	U1035	S-104	5-Day Academy	10/22/2025 14:30, 10/23/2025 13:15, 10/24/2025 09:55, 10/24/2025 11:20, 10/24/2025 16:10	10/22/2025 14:45, 10/23/2025 13:30, 10/24/2025 10:20, 10/24/2025 11:40, 10/24/2025 16:40	5	105
Marco Rossi	marco.rossi@example.com	U1036	S-104	5-Day Academy	10/22/2025 14:20, 10/23/2025 13:15, 10/23/2025 14:50	10/22/2025 15:00, 10/23/2025 14:30, 10/23/2025 15:35	3	160
Nadia Ansari	nadia.ansari@example.com	U1037	S-104	5-Day Academy	10/21/2025 10:55, 10/23/2025 09:50, 10/23/2025 15:30	10/21/2025 11:15, 10/23/2025 10:35, 10/23/2025 15:50	3	85
Omar Pillai	omar.pillai@example.com	U1038	S-104	5-Day Academy	10/21/2025 10:00, 10/22/2025 15:15, 10/23/2025 14:15, 10/24/2025 11:20, 10/24/2025 14:05	10/21/2025 10:30, 10/22/2025 15:55, 10/23/2025 15:30, 10/24/2025 11:50, 10/24/2025 15:20	5	250

Daily breakdown: missed time for each day separately

One row per learner per session day. Times within a day are comma-separated and in upload format (MM/DD/YYYY HH:MM).

Learner Name	Learner Email	User ID	Session ID	Program	Date	Day #	Missed Time Start	Missed Time End	Missed periods	Missed minutes
Ben Carter	ben.carter@example.com	U1002	S-101	1-Day Workshop	10/06/2025	1	10/06/2025 09:40, 10/06/2025 11:00	10/06/2025 09:55, 10/06/2025 12:15	2	90
Dev Patel	dev.patel@example.com	U1004	S-101	1-Day Workshop	10/06/2025	1	10/06/2025 09:45	10/06/2025 10:00	1	15
Jon Brooks	jon.brooks@example.com	U1009	S-101	1-Day Workshop	10/06/2025	1	10/06/2025 10:10, 10/06/2025 15:10	10/06/2025 11:25, 10/06/2025 15:25	2	90
Liam Murphy	liam.murphy@example.com	U1011	S-102	2-Day Bootcamp	10/08/2025	2	10/08/2025 10:50	10/08/2025 12:05	1	75
Olivia Reed	olivia.reed@example.com	U1014	S-102	2-Day Bootcamp	10/08/2025	2	10/08/2025 13:25	10/08/2025 14:25	1	60
Tariq Ali	tariq.ali@example.com	U1018	S-102	2-Day Bootcamp	10/07/2025	1	10/07/2025 10:30, 10/07/2025 14:50	10/07/2025 11:10, 10/07/2025 15:50	2	100
Tariq Ali	tariq.ali@example.com	U1018	S-102	2-Day Bootcamp	10/08/2025	2	10/08/2025 09:20	10/08/2025 09:40	1	20
Anil Verma	anil.verma@example.com	U1025	S-103	4-Day Program	10/15/2025	3	10/15/2025 11:55, 10/15/2025 13:10, 10/15/2025 15:40	10/15/2025 13:10, 10/15/2025 13:50, 10/15/2025 16:10	3	145
Anil Verma	anil.verma@example.com	U1025	S-103	4-Day Program	10/16/2025	4	10/16/2025 15:15	10/16/2025 15:40	1	25
Bianca Silva	bianca.silva@example.com	U1026	S-103	4-Day Program	10/13/2025	1	10/13/2025 14:00	10/13/2025 14:30	1	30
Carlos Nguyen	carlos.nguyen@example.com	U1027	S-103	4-Day Program	10/16/2025	4	10/16/2025 10:15	10/16/2025 11:15	1	60
Elena Ortiz	elena.ortiz@example.com	U1029	S-103	4-Day Program	10/13/2025	1	10/13/2025 13:10, 10/13/2025 15:55	10/13/2025 13:55, 10/13/2025 16:20	2	70
Elena Ortiz	elena.ortiz@example.com	U1029	S-103	4-Day Program	10/16/2025	4	10/16/2025 16:00	10/16/2025 16:20	1	20
Vikram Bose	vikram.bose@example.com	U1020	S-103	4-Day Program	10/13/2025	1	10/13/2025 10:10	10/13/2025 10:55	1	45
Vikram Bose	vikram.bose@example.com	U1020	S-103	4-Day Program	10/16/2025	4	10/16/2025 09:10	10/16/2025 09:35	1	25
Wen Chen	wen.chen@example.com	U1021	S-103	4-Day Program	10/14/2025	2	10/14/2025 09:40	10/14/2025 10:20	1	40
Xavier Fischer	xavier.fischer@example.com	U1022	S-103	4-Day Program	10/15/2025	3	10/15/2025 09:50, 10/15/2025 11:30, 10/15/2025 14:50	10/15/2025 10:35, 10/15/2025 12:15, 10/15/2025 16:05	3	165
Yara Ahmed	yara.ahmed@example.com	U1023	S-103	4-Day Program	10/13/2025	1	10/13/2025 09:45, 10/13/2025 13:25	10/13/2025 10:05, 10/13/2025 13:35	2	30
Zane Kapoor	zane.kapoor@example.com	U1024	S-103	4-Day Program	10/15/2025	3	10/15/2025 11:10	10/15/2025 11:25	1	15
Faisal Hassan	faisal.hassan@example.com	U1030	S-104	5-Day Academy	10/20/2025	1	10/20/2025 09:40, 10/20/2025 13:10	10/20/2025 10:00, 10/20/2025 13:55	2	65
Faisal Hassan	faisal.hassan@example.com	U1030	S-104	5-Day Academy	10/21/2025	2	10/21/2025 13:45	10/21/2025 14:25	1	40
Faisal Hassan	faisal.hassan@example.com	U1030	S-104	5-Day Academy	10/22/2025	3	10/22/2025 11:40, 10/22/2025 13:40	10/22/2025 12:00, 10/22/2025 14:55	2	95
Grace Park	grace.park@example.com	U1031	S-104	5-Day Academy	10/20/2025	1	10/20/2025 13:05	10/20/2025 14:20	1	75
Harsh Malik	harsh.malik@example.com	U1032	S-104	5-Day Academy	10/20/2025	1	10/20/2025 09:05	10/20/2025 10:20	1	75
Kiran Bhat	kiran.bhat@example.com	U1034	S-104	5-Day Academy	10/21/2025	2	10/21/2025 11:35, 10/21/2025 14:40, 10/21/2025 16:15	10/21/2025 11:55, 10/21/2025 15:25, 10/21/2025 16:35	3	85
Kiran Bhat	kiran.bhat@example.com	U1034	S-104	5-Day Academy	10/22/2025	3	10/22/2025 09:40, 10/22/2025 12:00	10/22/2025 10:40, 10/22/2025 12:45	2	105
Kiran Bhat	kiran.bhat@example.com	U1034	S-104	5-Day Academy	10/24/2025	5	10/24/2025 10:00, 10/24/2025 14:05, 10/24/2025 15:50	10/24/2025 10:20, 10/24/2025 14:20, 10/24/2025 16:20	3	65
Leela Hughes	leela.hughes@example.com	U1035	S-104	5-Day Academy	10/22/2025	3	10/22/2025 14:30	10/22/2025 14:45	1	15
Leela Hughes	leela.hughes@example.com	U1035	S-104	5-Day Academy	10/23/2025	4	10/23/2025 13:15	10/23/2025 13:30	1	15
Leela Hughes	leela.hughes@example.com	U1035	S-104	5-Day Academy	10/24/2025	5	10/24/2025 09:55, 10/24/2025 11:20, 10/24/2025 16:10	10/24/2025 10:20, 10/24/2025 11:40, 10/24/2025 16:40	3	75
Marco Rossi	marco.rossi@example.com	U1036	S-104	5-Day Academy	10/22/2025	3	10/22/2025 14:20	10/22/2025 15:00	1	40
Marco Rossi	marco.rossi@example.com	U1036	S-104	5-Day Academy	10/23/2025	4	10/23/2025 13:15, 10/23/2025 14:50	10/23/2025 14:30, 10/23/2025 15:35	2	120
Nadia Ansari	nadia.ansari@example.com	U1037	S-104	5-Day Academy	10/21/2025	2	10/21/2025 10:55	10/21/2025 11:15	1	20
Nadia Ansari	nadia.ansari@example.com	U1037	S-104	5-Day Academy	10/23/2025	4	10/23/2025 09:50, 10/23/2025 15:30	10/23/2025 10:35, 10/23/2025 15:50	2	65

Omar Pillai	omar.pillai@example.com	U1038	S-104	5-Day Academy	10/21/2025	2	10/21/2025 10:00	10/21/2025 10:30	1	30
Omar Pillai	omar.pillai@example.com	U1038	S-104	5-Day Academy	10/22/2025	3	10/22/2025 15:15	10/22/2025 15:55	1	40
Omar Pillai	omar.pillai@example.com	U1038	S-104	5-Day Academy	10/23/2025	4	10/23/2025 14:15	10/23/2025 15:30	1	75
Omar Pillai	omar.pillai@example.com	U1038	S-104	5-Day Academy	10/24/2025	5	10/24/2025 11:20, 10/24/2025 14:05	10/24/2025 11:50, 10/24/2025 15:20	2	105

Daily matrix: missed start and end time for every day, side by side

Times shown as HH:MM. A dash means no missed time that day; grey cells are days outside the learner's program.

Learner Name	Session ID	Program	Day 1 Start	Day 1 End	Day 2 Start	Day 2 End	Day 3 Start	Day 3 End	Day 4 Start	Day 4 End	Day 5 Start	Day 5 End	Total instances	Total minutes
Ben Carter	S-101	1-Day Workshop	09:40, 11:00	09:55, 12:15									2	90
Dev Patel	S-101	1-Day Workshop	09:45	10:00									1	15
Jon Brooks	S-101	1-Day Workshop	10:10, 15:10	11:25, 15:25									2	90
Liam Murphy	S-102	2-Day Bootcamp	–	–	10:50	12:05							1	75
Olivia Reed	S-102	2-Day Bootcamp	–	–	13:25	14:25							1	60
Tariq Ali	S-102	2-Day Bootcamp	10:30, 14:50	11:10, 15:50	09:20	09:40							3	120
Anil Verma	S-103	4-Day Program	–	–	–	–	11:55, 13:10, 15:40	13:10, 13:50, 16:10	15:15	15:40			4	170
Bianca Silva	S-103	4-Day Program	14:00	14:30	–	–	–	–	–	–			1	30
Carlos Nguyen	S-103	4-Day Program	–	–	–	–	–	–	10:15	11:15			1	60
Elena Ortiz	S-103	4-Day Program	13:10, 15:55	13:55, 16:20	–	–	–	–	16:00	16:20			3	90
Vikram Bose	S-103	4-Day Program	10:10	10:55	–	–	–	–	09:10	09:35			2	70
Wen Chen	S-103	4-Day Program	–	–	09:40	10:20	–	–	–	–			1	40
Xavier Fischer	S-103	4-Day Program	–	–	–	–	09:50, 11:30, 14:50	10:35, 12:15, 16:05	–	–			3	165
Yara Ahmed	S-103	4-Day Program	09:45, 13:25	10:05, 13:35	–	–	–	–	–	–			2	30
Zane Kapoor	S-103	4-Day Program	–	–	–	–	11:10	11:25	–	–			1	15
Faisal Hassan	S-104	5-Day Academy	09:40, 13:10	10:00, 13:55	13:45	14:25	11:40, 13:40	12:00, 14:55	–	–	–	–	5	200
Grace Park	S-104	5-Day Academy	13:05	14:20	–	–	–	–	–	–	–	–	1	75
Harsh Malik	S-104	5-Day Academy	09:05	10:20	–	–	–	–	–	–	–	–	1	75
Kiran Bhat	S-104	5-Day Academy	–	–	11:35, 14:40, 16:15	11:55, 15:25, 16:35	09:40, 12:00	10:40, 12:45	–	–	10:00, 14:05, 15:50	10:20, 14:20, 16:20	8	255
Leela Hughes	S-104	5-Day Academy	–	–	–	–	14:30	14:45	13:15	13:30	09:55, 11:20, 16:10	10:20, 11:40, 16:40	5	105

Marco Rossi	S-104	5-Day Academy	–	–	–	–	14:20	15:00	13:15, 14:50	14:30, 15:35	–	–	3	160
Nadia Ansari	S-104	5-Day Academy	–	–	10:55	11:15	–	–	09:50, 15:30	10:35, 15:50	–	–	3	85
Omar Pillai	S-104	5-Day Academy	–	–	10:00	10:30	15:15	15:55	14:15	15:30	11:20, 14:05	11:50, 15:20	5	250

Impact: manual vs automated

Time per run

Measure	Value	Unit / Note
Manual processing time	30	minutes (approx., provided by project owner)
Automated processing time	30	seconds (approx., provided by project owner)
Automated time in minutes	0.5	minutes
Time saved per run	29.5	minutes
Reduction in processing time	**98.3%**	percent

Blue cells are inputs. The time figures describe the original form-processing task as reported by the project owner.